Cleaner Water 2 (2024) 100041

Contents lists available at ScienceDirect

Cleaner Water

journal homepage: www.sciencedirect.com/journal/cleaner-water

Bio-inspired MXene membranes for enhanced separation and anti-fouling in oil-in-water emulsions: SHAP explainability ML

Nadeem Baig [a], Sani I. Abba [b,c,*], Jamil Usman [a], Ibrahim Muhammad [d], Ismail Abdulazeez [a], A.G. Usman [e], Isam H. Aljundi [a,f]

[a] Interdisciplinary Research Center for Membrane and Water Security, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia
[b] Department of Chemical Engineering, Prince Mohammad Bin Fahd University, Al Khobar, 31952, Saudi Arabia
[c] Water Research Centre, Prince Mohammad Bin Fahd University, Al Khobar, 31952, Saudi Arabia
[d] Department of Pure and Industrial Chemistry, Faculty of Science, Sokoto State University, P.M.B. 2134, Along Birnin Kebbi Road, Sokoto, Nigeria
[e] Operational Research Centre in Healthcare, Near East University, Nicosia, Cyprus
[f] Department of Chemical Engineering, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

ARTICLE INFO

Keywords:
environmental sustainability
clean water access
machine learning
MXene
reliability analysis
SHAP

ABSTRACT

Optimizing membrane performance for efficient water treatment is crucial for sustainable development and environmental protection, aligning with UN SDGs. This study involves experimental design, statistical reliability of small data, and explainable machine learning (ML) using SHAP (Shapley Additive Explanations). The research uses ML models and statistical tests to ensure data reliability and stationarity and investigate various membranes' fouling and separation efficiency (MX-CM, PDMX-CM, and SPDMX-CM). Stationarity tests, including the Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) tests, revealed that MX-CM is stationary at level (I(0)), while PDMX-CM and SPDMX-CM required first differencing (I(1)) to achieve stationarity. SHAP analysis showed that in the fouling study, higher values of PDMX-CM and MX-CM positively influenced model predictions, with SHAP values of +0.09 for Cycle, −0.06 for PDMX-CM, and −0.06 for MX-CM. In the separation efficiency study, Cycle had a neutral impact (0.00), PDMX-CM had a slight positive effect, and MX-CM had a slight negative impact. These findings highlight the importance of ensuring data stationarity and utilizing SHAP for model explainability in predicting membrane performance. Accurate preprocessing and model interpretation enhance decision-making and optimization in membrane fouling and separation efficiency studies, ensuring robust and reliable ML models.

1. Introduction

As the demand for environmental sustainability and various industrial processes continues to rise, removing contaminants from water or separating mixtures into highly pure substances becomes increasingly crucial. The separation processes conducted at an industrial scale are usually based on high-energy demanding processes (Liu et al., 2020), whereas membrane-based technologies are considered more economically friendly. These promising technologies, including desalination, produced water treatment, and organics removal, can potentially revolutionize water applications. However, the critical challenge associated with these membranes is fouling, which is a persistent issue that leads to

Abbreviations: ML, Machine Learning; SHAP, Shapley Additive Explanations; ADF, Augmented Dickey-Fuller; PP, Phillips-Perron; SPDMX-CM, Sulfonated Polydopamine Functionalized MXene Ceramic Membrane; PDMX-CM, Polydopamine Functionalized MXene Ceramic Membrane; LMH, Liters per Meter Squared per Hour; EPA, Environmental Protection Agency; SDGs, Sustainable Development Goals; PDA, Polydopamine; HF, Hydrofluoric Acid; SD, Standard Deviation; ACF, Autocorrelation Function; PACF, Partial Autocorrelation Function; KPSS, Kwiatkowski-Phillips-Schmidt-Shin; ARIMA, Autoregressive Integrated Moving Average; MOF, Metal-Organic Framework; AIC, Akaike Information Criterion; BK, Bartlett Kernel; SC, Schwarz Criterion; TV, Test Value; CV, Critical Value; TT, Test Type; ICC, Intraclass Correlation Coefficient; KR-20, Kuder-Richardson Formula 20; EDA, Exploratory Data Analysis; FRR, Flux Recovery Ratio; TiO2, Titanium Dioxide; 2D, Two-Dimensional; GO, Graphene Oxide; XAI, Explainable Artificial Intelligence; LIME, Local Interpretable Model-Agnostic Explanations; TVET, Technical and Vocational Education and Training; ANOVA, Analysis of Variance; PLS, Partial Least Squares; Cf, concentration in the feed; Cp, concentration of the permeate; V, permeate volume; T, time; F, flux.

* Corresponding author at: Department of Chemical Engineering, Prince Mohammad Bin Fahd University, Al Khobar, 31952, Saudi Arabia.
E-mail addresses: saniisaabba86@gmail.com, sani.abba@kfupm.edu.sa (S.I. Abba).

https://doi.org/10.1016/j.clwat.2024.100041
Received 16 July 2024; Received in revised form 31 August 2024; Accepted 9 September 2024
Available online 13 September 2024
2950-2632/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

a decrease in solvent permeation over time (Tummons et al., 2020) (Sutrisna et al., 2022). To address this, two-dimensional materials have garnered significant attention for designing the next-generation membranes, offering improved fouling resistance (Baig, 2023). The atomic level thickness, excellent flexibility, high mechanical stretch, and potential for large-scale production make these materials theoretically ideal for membrane applications (Ying et al., 2016). The membranes rationally designed with two-dimensional materials can offer high selectivity, permeability, and fouling resistance. The two-dimensional materials with atomic thickness can be used to design ultrathin membranes, offering an unprecedented permeation rate (Cheng et al., 2021). Several two-dimensional membranes have been designed for precise molecular sieving (Zhang et al., 2018) (Chen and Yam, 2020). Further exploration of utilizing 2D materials for membrane fabrication is ongoing.

Graphene is the profound star of the 2D family that has been extensively explored as a membrane material (Baig et al., 2023a). Graphene has certain limitation, as the graphene itself is hydrophobic in nature and lack the functional groups which makes it tough to disperse or process in the solution to design the membranes. It is the reason that graphene oxide which is the derivative of the graphene is usually used for membrane fabrications (Al-Hamadani et al., 2020). Recently, MXene has been rapidly emerging as a new class of 2D materials and is being investigated for use in designing advanced membranes (Al-Hamadani et al., 2020). The MXene is typically represented by the general formula Mn+1XnTx, wherein M denotes the early transition metals, X signifies the presence of carbides, nitrides, or both, and T indicates the presence of terminal groups (Kwon et al., 2022). A major challenge is associated with the stability of MXene sheets. MXene have contained several oxygen containing hydrophilic groups which provided the further opportunity to functionalize or tune the 2D sheets (Chertopalov and Mochalin, 2018b). However, it has been observed that in the colloidal state, the MXene flakes cannot resist oxidation. The colloidal solution of the MXene turns from black to white as the MXene oxidizes to TiO_2 (Chertopalov and Mochalin, 2018a). The formation of the TiO_2 has been started from the edges of the MXene sheets, and ultimately, the structure of the MXene collapsed. Therefore, finding an effective way to enhance the stability of the MXene sheets has become extremely critical (Chertopalov and Mochalin, 2018b). Several pieces of literature have reported that MXene functionalization can enhance the stability of MXene sheets (Kim et al., 2021) Utilization of MXene as membrane material, apart from oxidative degradation, its stability on the membrane surface to form a stable separating layer is also a critical challenge. Secondly, the fouling resistance of the pristine MXene sheets is not up to the mark to make them effective for oil/water separation. Therefore, the appropriate functionalization can assist in forming the antifouling MXene membrane, and the stabilizing agent enhances the stability on the surface of the support membrane. In a few examples, machine learning (ML) has been used for MXene.

For instance, ML has proven revolutionary in speeding up scientific discoveries in highly demanding areas. ML models, once trained, can swiftly and accurately predict outcomes by uncovering nonlinear relationships between input features and their corresponding characteristics (Cao et al., 2023) (Usman et al., 2024). The utilization of ML can evaluate the broad spectrum of the membranes, which can significantly impact the membranes' optimization time and, at the same time, less utilization of the computational and experimental resources to evaluate the membrane performances. The 2D materials are experimentally and computationally extensively explored (Baig et al., 2023b) (Wei et al., 2019). The computational model is usually applied to known structures, and it's a fact that the 2D materials can be evaluated experimentally, which can be synthesized in the lab. Therefore, significant interest has recently increased in utilizing ML to predict or screen 2D materials for advanced membrane fabrications (Paechotrattanakul et al., 2023; Priya et al., 2022). As discussed, MXene is rapidly emerging as a new 2D material with great potential for designing advanced oil/water separation membranes.

However, ML has been utilized to predict MXene synthesis (Frey et al., 2019a) and its electrochemical and energy storage properties (Qian et al., 2022). This study aims to evaluate and optimize the fouling and separation efficiency of various membranes (MX-CM, PDMX-CM, and SPDMX-CM) using statistical analyses and advanced ML models. For this purpose, reliability and stationarity data-driven analysis, including Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests, were conducted to assess and ensure experimental model reliability, implementing preprocessing steps like differencing to address non-stationarity and utilizing Shapley Additive exPlanations (SHAP) to interpret feature impacts on model predictions. The study also aims to enhance ML model transparency by identifying features affecting fouling and separation efficiency, thereby optimizing membrane performance in water treatment applications. This contributes to Sustainable Development Goals (SDGs) and aligns with Environmental Protection Agency (EPA) standards for environmental sustainability. Ultimately, the study demonstrates the effectiveness of integrating statistical tests and ML explainability tools in ensuring data reliability and enhancing model transparency, providing a validated framework for future membrane performance studies. These membranes separate the surfactant-stabilized oily waster emulsion (Baig et al., 2023c).

The motivation for the current study derives from the increasing recognition of the power of small data in chemistry, particularly in the context of experimental data. In the field of membrane technology for water treatment, acquiring large datasets can be challenging due to the high complexity and cost of experiments. Small data, characterized by high-quality, detailed datasets, provides a unique opportunity to extract meaningful insights and achieve reliable modeling outcomes without extensive data collection efforts. However, harnessing small data allows for a deeper understanding of membrane performance, as each data point is rich with information and accurately collected under controlled experimental conditions. This approach aligns with the principles of precision and accuracy, which are paramount in chemical research and engineering. By focusing on small, high-quality datasets, researchers can develop robust and interpretable ML models that can effectively predict membranes' fouling and separation efficiency. Moreover, using advanced statistical tests like the ADF and PP tests ensures the reliability and stationarity of these small datasets, which is crucial for accurate time series modeling. The integration of SHAP further enhances model transparency, allowing researchers to understand the impact of individual features on model predictions, thereby facilitating better decision-making and optimization.

2. Methods and methodology

2.1. Synthesis of the sulofnated the MXene

The MXene sheets were synthesized by etching the MAX phase with the HF. The modification of the MXene with the sulfonated groups consists of several steps. For delamination of the MXene sheets and to get the uniform dispersion of its sheets, the MXene was sonicated for 2 h. After delimitation of the MXene sheets, the dopamine was added into the MXene solution, and the pH was increased to 8.5, which resulted in the initiation of the polymerization of the dopamine. The solution prepared for membrane fabrication was consist of dopamine (0.5 mg/ML) and MXene (1 mg/ML). The reaction mixture was kept at 60 °C for 24 h to enhance the polymerization. The pristine MXene solution was black in color, and it turned to dark brownish black. For sulfonation, the 2,5-diamino benzenesulfonic acid was added to the MXene and dopamine solution during the polymerization of the dopamine. This results in the formation of sulfonated MXene. The 2,5-diamino benzenesulfonic acid concentration of 0.1 mg/ML were used to react with the polydopamine. Further, details of the sulfonation of the MXene have already been discussed in the previously published literature (Baig et al., 2023c).

2.2. Designing sulfonated MXene membrane

The sulfonated MXene separating layer was prepared on the surface of the ceramic membrane. A 1 mg/ML solution of the MXene was prepared in double distilled water. To form the separating layer, 25 ML was taken from the stock solution, diluted 10 times, and subjected to sonication to separate the sheets further. The dead-end filtration assembly was used to form the separating layer. Therefore, the ceramic membrane was installed into the separation assembly. The sulfonated polydopamine functionalized MXene was added into the dead-end cell and slowly passed at a pressure of 0.5 bar, where sulfonated polydopamine functionalized MXene kept self-assembling to form an MXene-based separating layer on the surface of the ceramic-membrane. After forming the barrier layer, it was compacted at a 1 and 2 bar pressure, respectively. MX-CM represents the membranes which only have the pristine MXene as separating layer on the surface of the ceramic membrane whereas the PDMX-CM membranes have s separating layer of the polydopamine functionalized MXene sheets. The SPDMX-CM represents the membranes in which the separating layer is formed by using the Sulfonated polydopamine functionalized MXene (Anjum et al., 2022). The surface morphology of the SPDMX-CM membrane can be seen in Fig. S1.

2.3. Preparation of oil-in-water emulsions

The surfactant-stabilized oil-in-water emulsion was prepared with the help of the tween-80. In brief, 0.2 g of diesel oil was added to 1 L of water, which contained 0.5 g of the surfactant. The stirring was done at 800 rpm for 24 hours. After that, the milky colour oil-in-water emulsion was formed.

2.4. Sulfonated polydopamine functionalized MXene membrane performance evaluation

The sulfonated polydopamine-functionalized MXene membrane was installed into the dead-end cell to assess its evaluation and fouling resistance. The membranes' permeation flux was measured using Eq. (1).

$$F = V/(A \times t) \quad (1)$$

where F is the flux (LMH) of membranes, V is the permeate volume (L), A is the membrane's area (m2), and t is the time (h) to collect the permeate volume. The membranes' separation efficiency was measured using Eq. (2).

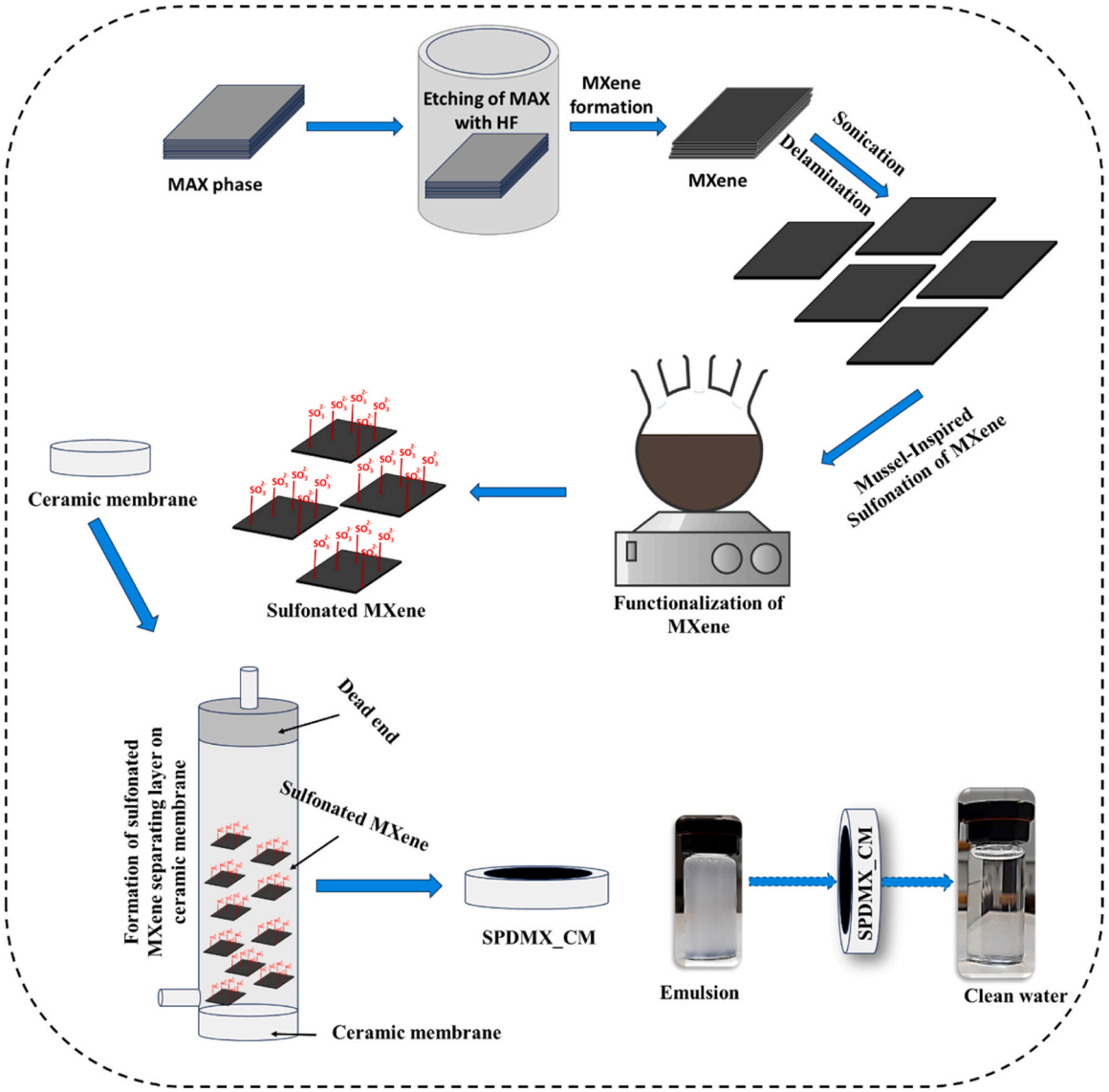


Fig. 1. Experimental workflow diagram.

$$E = \left(1 - \frac{Cf}{Cp}\right) \times 100 \tag{2}$$

Where E (%) is the separation efficiency, Cf is the oil concentration in the feed, and Cp is the concentration of the permeate. The flux recovery ratio of the membranes was found using Eq. (3).

$$FRR(\%) = \left(\frac{F2}{F1}\right) \times 100 \tag{3}$$

Where FRR is the flux recovery ratio, F2 is the recovery to the initial flux after cleaning the fouled membranes, and F1 is the initial flux of the membrane before fouling. (Fig. 1).

3. Proposed data-driven methods

3.1. Explainability AI (SHapley Additive exPlanations (SHAP))

ML models often achieve exceptional performance in predictive tasks but tend to lack interpretability, rendering them "black boxes". Understanding the complex nonlinear interactions between features within these models is crucial. SHAP address this issue by providing a means to interpret and understand model behavior and the relationships between features. SHAP values, derived from game theory concepts introduced by Lundberg and Lee, quantify the contribution of each feature to the model's predictions (Lu et al., 2023; Anjum et al., 2022). These values can indicate both positive and negative impacts on the output. This study employs various SHAP plots, including feature importance, summary plots, force plots, decision plots, and feature dependence plots, to elucidate the model's inner workings (Lundberg and Lee, 2017). By leveraging SHAP values, researchers can assess the significance of feature contributions, enhancing the transparency and interpretability of complex ML models (Lehnert et al., 2024).

Explainable ML is crucial for understanding and improving the reliability analysis of separation efficiency and anti-fouling performance in sulfonated polydopamine functionalized MXene-based membranes within surfactant-stabilized oil-in-water emulsions (Arrieta et al., 2024). By enhancing interpretability, transparency, error diagnosis, and regulatory compliance, explainable ML enables researchers to distinguish how variables such as membrane properties, surfactant concentration, and emulsion characteristics impact performance outcomes. Techniques like feature importance ranking, partial dependence plots, SHAP values, and LIME facilitate this interpretability (Rodríguez-Pérez and Bajorath, 2020; Yan et al., 2020). These methods explain the decision-making processes of complex models, building trust and aiding in error rectification. In this context, data collection and preprocessing, model building, and the application of explainability techniques are critical steps. Through these processes, explainable ML bridges the gap between high-performance predictive models and the need for transparent, reliable insights in scientific research and industrial applications, ultimately enhancing the trustworthiness and robustness of ML models used in the analysis of advanced materials and processes (Iban, 2022).

The function of the model, g(x') is defined as Eq. (4)

$$g(x') = \phi_0 + \sum_{i=1}^{M} \phi_i x' \tag{4}$$

Where x' is the generalized input variables in vector form.M is the number of features in the set. $\phi_0$and $\phi_i$are the attribution value of each variable used in the study.

The additive feature is generally attributed to desired properties, mainly as local accuracies, missingness, and consistency. If these properties are inhibited, a better explanation model g(x) can be achieved with a unique solution which is explained by Eq. (5).

$$\phi_i(f, x) = \frac{1}{N!} \sum_{S \subseteq P \setminus x_i} [|S|!(N - |S| - 1)!][f(S \cup \{x_i\}) - f(S)] \tag{5}$$

Where, S is the feature subset used, x is the vector space of feature values to be explained, $f(S)$refers to the model output value for the specific feature combination.

3.2. Concept of data reliability, normality and stationarity

Data reliability is crucial for ensuring the consistency and accuracy of data, which underpins the validity and dependability of results in modeling processes. Ensuring data reliability, normality, and stationarity enhances model performance, interpretability, and generalizability, which is essential for accurate predictions and robust conclusions in various fields, including chemistry, material science, environmental science, and material studies. To ensure data reliability and assess stationarity in datasets, various statistical methods are employed. For data reliability, methods like Cronbach's Alpha measure internal consistency, Intraclass Correlation Coefficient (ICC) assesses measurement reliability, Bland-Altman Analysis evaluates agreement between different measurement techniques, and Kuder-Richardson Formula 20 (KR-20) is used for dichotomous items (Hamisu et al., 2017). Normality tests such as the Shapiro-Wilk Test, Kolmogorov-Smirnov Test, Anderson-Darling Test, and Lilliefors Test verify if data follows a normal distribution (Mati, 2021). Stationarity tests like the ADF Test, Phillips-Perron (PP) Test, Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test, and Zivot-Andrews Test check for unit roots and structural breaks in time series data. Homogeneity of variance tests, including Levene's Test, Bartlett's Test, and Brown-Forsythe Test, assess the equality of variances across groups. Additionally, test-retest reliability and split-half reliability measure stability and consistency over time. Robustness and outlier detection methods like Mahalanobis Distance and Cook's Distance identify influential data points. These comprehensive statistical methods are critical in ensuring the accuracy and reliability of predictive models, particularly in fields like environmental science where data quality directly impacts the validity of conclusions (Shuabiu et al., 2021). In ML and statistical modelling, reliable data significantly affects the performance, interpretability, and generalizability of models (Yu et al., 2022).

Further numerical analysis was conducted using correlograms, which include the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), which is essential in assessing data reliability and enhancing modeling in time series analysis. The ACF measures the correlation between observations at different lags, helping to identify patterns, seasonality, and trends, while the PACF measures the direct correlation after accounting for earlier lags, assisting in the specification of autoregressive models (Wilson, 2016). These functions are crucial for determining the appropriate parameters for ARIMA models and ensuring correct model specification. Additionally, ACF and PACF plots help detect stationarity, a key assumption in time series modeling, indicating whether data transformations like differencing are needed. After fitting a model, examining the ACF and PACF of residuals helps diagnose model adequacy, ensuring that residuals resemble white noise. This diagnostic step is vital for refining models to improve accuracy. By identifying and addressing issues such as seasonality, trends, and correlations through ACF and PACF, the quality of data preprocessing is enhanced, leading to more reliable and accurate predictive models (Wilson, 2016).

3.3. Small data handling

The ML algorithms, known for the data-driven black-box approach, have been successfully used to map trends in chemistry and materials science without needing prior knowledge of underlying physical mechanisms. However, a significant challenge in material science is the

acquisition of small datasets due to the high complexity of experimental procedures (Frey et al., 2019b). Researchers often prefer to collect smaller, high-quality samples in controlled environments rather than relying on larger datasets with uncertain origins. This preference highlights the need to distinguish between "big data" and "small data" within the fields of computer science and material science. Xu et al. (2023). noted that most datasets used in material science ML are categorized as small data (Xu et al., 2023). Despite this, the predictive goals of ML models remain consistent, requiring careful handling of small datasets due to increased uncertainty compared to larger datasets. Challenges such as underfitting, overfitting, and data imbalance are more pronounced with small datasets, making accurate data preprocessing and post-processing essential. Selecting appropriate modeling algorithms is also critical to ensure the reliability and accuracy of predictions. By prioritizing these steps, researchers can mitigate the inherent challenges of working with small data in material science ML, thereby enhancing the effectiveness of their models (Liang and Zhu, 2019).

The "Small data" refers to datasets that are small in size but rich in information content, focusing on quality and depth rather than volume. Small data is powerful in modeling due to its ability to provide focused insights, its manageability, and its efficiency, requiring less computational power (Fawcett, 2019). Leveraging small data involves maximizing the value of limited datasets through careful data preprocessing, feature engineering, and applying robust statistical and ML methods designed for small samples. The objective is to extract meaningful patterns and make reliable predictions despite the smaller dataset size. This approach allows for precise, actionable insights, reduced risk of errors and biases, and efficient model performance (Jahani et al., 2023). Fig. 2a outlines the workflows and the general process of experimental and modern-based ML-assisted materials design. As illustrated in Fig. 2b, the approaches to handling small data are discussed at three levels. At the data source level, strategies include data extraction from publications, constructing materials databases, and employing high-throughput computations and experiments. At the algorithm level, methods focus on modelling algorithms suitable for small data and techniques for imbalanced learning.

4. Results and discussion

4.1. Pre-analysis of the experimental data

Exploratory Data Analysis (EDA) and data preprocessing are essential for understanding and preparing data for analysis. EDA involves summarizing data characteristics using descriptive statistics and

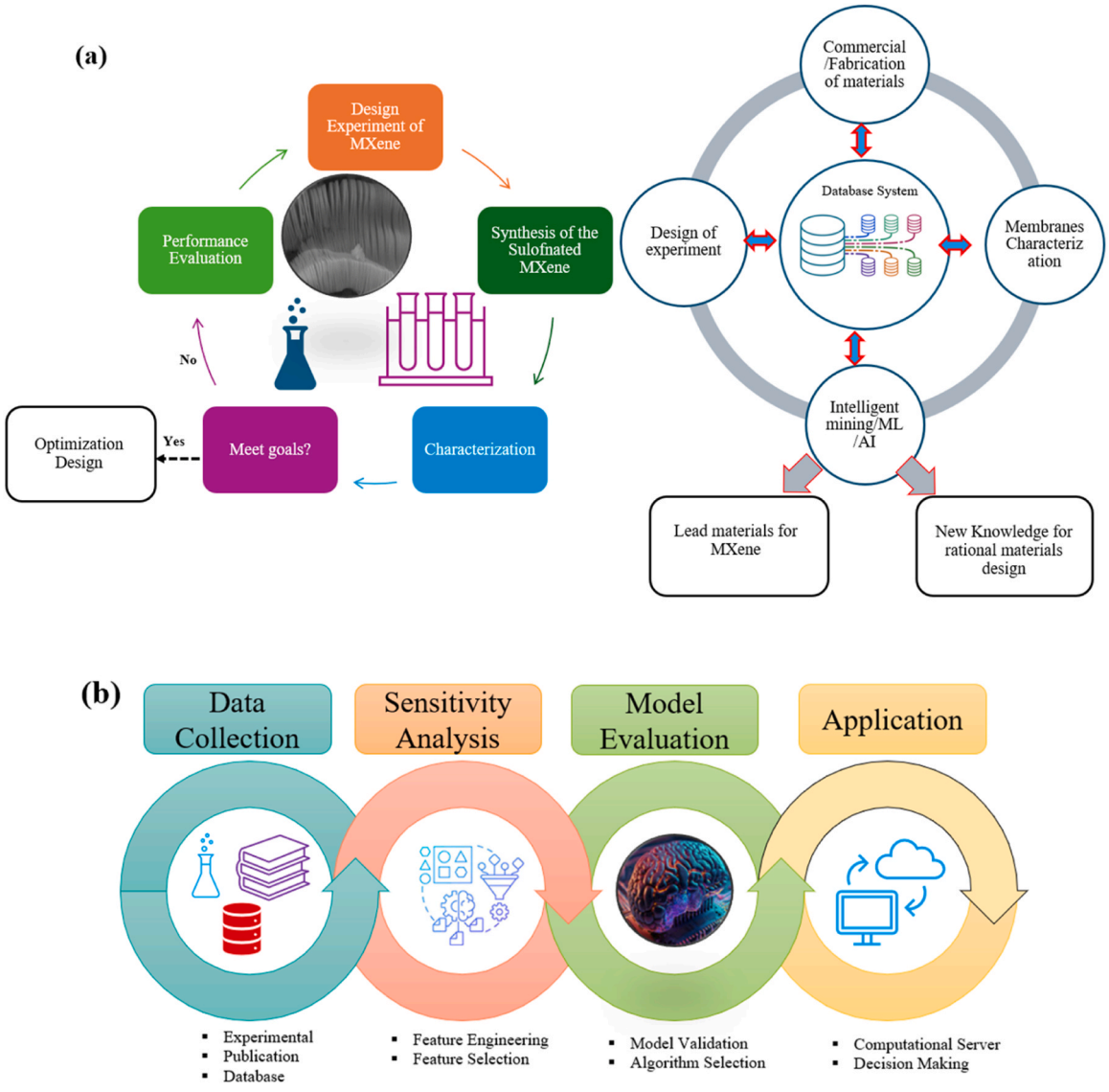


Fig. 2. (a) Schematic of the traditional experimental design process and schematic of ML integrated experimental scheme (b) ML model steps.

visualizations, helping to uncover patterns, variances, and relationships. These steps enhance data reliability, meet model assumptions, and improve the accuracy and robustness of subsequent analyses, making EDA and preprocessing critical for effective data-driven decision-making. For this purpose, Table 1 provides descriptive statistics for a fouling study, examining the performance of various membrane types (MX-CM, PDMX-CM, and SPDMX-CM) over multiple cycles. The mean number of cycles is 8 with a standard deviation (SD) of 4.47, showing significant variability. For the MX-CM membrane, the mean fouling resistance is 0.277 with an SD of 0.248 and a skewness of 2.24, indicating moderate performance with high variability and a right-skewed distribution. The PDMX-CM membrane shows better performance with a mean of 0.453, an SD of 0.167, and a skewness of 2.73, though it also displays high variability and a skewed distribution. The SPDMX-CM membrane has the highest mean fouling resistance at 0.816, the lowest SD at 0.130, and a skewness of −0.488, suggesting the best and most consistent performance with a slight left-skewed distribution.

Similarly, the descriptive statistics for a separation efficiency study is presented in Table 1, evaluating the performance of different membrane types (MX-CM, PDMX-CM, and SPDMX-CM) over multiple cycles. The mean number of cycles is 8 with a standard deviation (SD) of 4.47, indicating significant variability. For the MX-CM membrane, the mean separation efficiency is 92.145 with an SD of 1.792 and a skewness of −0.626, suggesting good performance with moderate variability and a slightly left-skewed distribution. The PDMX-CM membrane shows a mean efficiency of 93.226, an SD of 1.520, and a skewness of −0.788, indicating slightly better performance with less variability and a more pronounced left-skew. The SPDMX-CM membrane exhibits the highest mean efficiency at 95.913, the lowest SD at 0.928, and a skewness of −0.255, reflecting the best and most consistent performance with minimal skew. In general, SPDMX-CM appears to be the most reliable and effective membrane in terms of fouling resistance and separation efficiency.

4.2. Dependency analysis results

Fig. 3 presents the dependency for a fouling study involving different membrane types (MX-CM, PDMX-CM, and SPDMX-CM) and the number of cycles. The number of cycles shows strong negative correlations with all three membranes, indicating that fouling performance decreases as cycles increase: −0.759 for MX-CM, −0.756 for PDMX-CM, and −0.957 for SPDMX-CM. The positive correlations between the membranes suggest similar fouling behaviour, with MX-CM and PDMX-CM having the highest positive correlation at 0.927, indicating closely related performance. SPDMX-CM shows strong positive correlations of 0.711 with MX-CM and 0.733 with PDMX-CM. These relationships help in predicting and optimizing membrane performance under varying conditions. On the other hand, for separation efficiency study involving different membrane types (MX-CM, PDMX-CM, and SPDMX-CM) and the number of cycles. The number of cycles shows strong negative correlations with the separation efficiency of all three membranes, indicating that efficiency decreases as cycles increase: −0.970 for MX-CM, −0.960 for PDMX-CM, and −0.977 for SPDMX-CM. The positive dependency between the membranes suggest similar efficiency behaviours, with MX-CM and PDMX-CM having the highest positive correlation at 0.981, indicating closely related performance. SPDMX-CM shows strong positive correlations of 0.964 with MX-CM and 0.955 with PDMX-CM (Fig. 3). These relationships highlight that while all membranes experience a reduction in separation efficiency with increased cycles, their performances are strongly interrelated, helping in predicting and optimizing membrane efficiency under varying operational conditions.

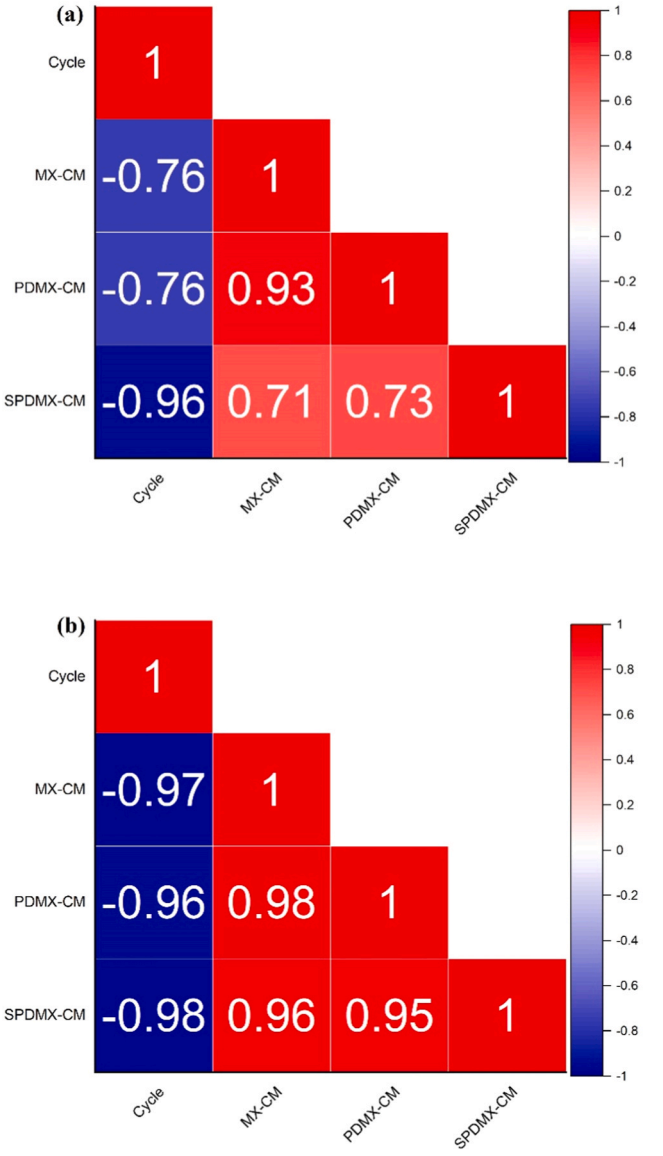


Fig. 3. Dependency analysis between different membranes and cycle time for fouling and separation efficiency.

Furthermore, the correlograms of fouling study including the ACF and PACF are vital for data reliability and modeling in time series analysis. This process enhances data preprocessing, leading to more reliable and accurate predictive models (see, Fig. 4a). The correlogram of SPDMX_CM shows significant ACF and PACF for 12 lags from 15 observations. The first three lags have high autocorrelation values of 0.729, 0.433, and 0.253, respectively, indicating strong correlations

Table 1
Descriptive statistic of fouling and separation efficiency study.

	Fouling Study			
	Cycle	*MX-CM*	*PDMX-CM*	*SPDMX-CM*
Mean	8.000	0.277	0.453	0.816
SD	4.472	0.248	0.167	0.130
Kurtosis	−1.200	4.827	9.202	−0.046
Skewness	0.000	2.236	2.728	−0.488
Minimum	1.000	0.121	0.264	0.563
Maximum	15.000	1.000	1.000	1.000
	Separation Efficiency Study			
	Cycle	*MX-CM*	*PDMX-CM*	*SPDMX-CM*
Mean	8.000	92.145	93.226	95.913
SD	4.472	1.792	1.520	0.928
Kurtosis	−1.200	−0.543	−0.273	−0.593
Skewness	0.000	−0.626	−0.788	−0.255
Minimum	1.000	88.800	90.032	94.080
Maximum	15.000	94.400	94.864	97.272

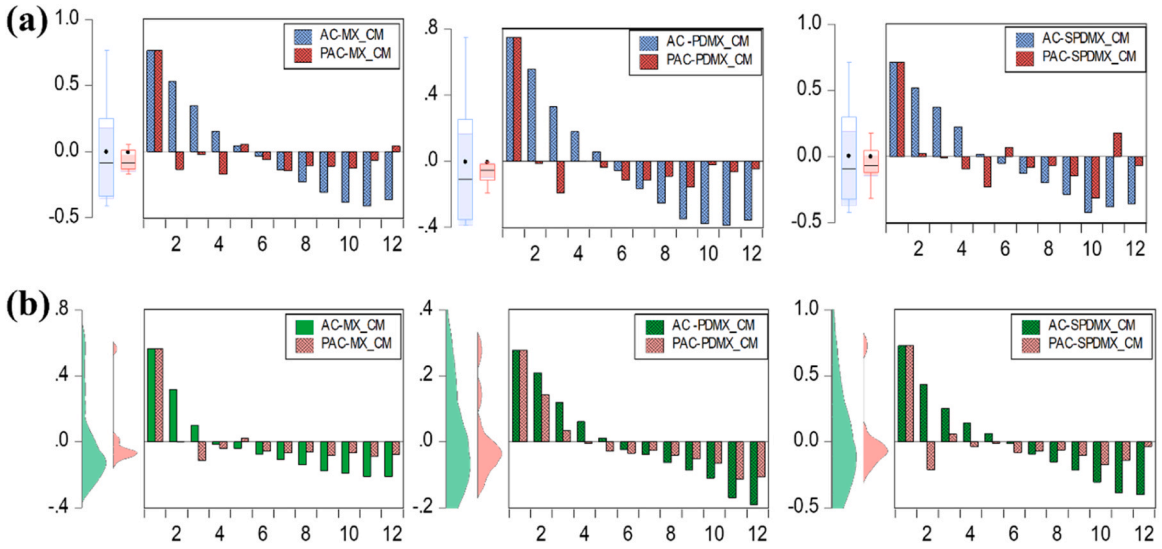


Fig. 4. Correlograms based on ACF and PACF for (a) fouling study (b) separation efficiency.

between data points and their preceding values, while negative correlations appear at lag 8 (-0.154) and become more pronounced at lags 9 (-0.216), 10 (-0.306), 11 (-0.390), and 12 (-0.401). The PACF shows a high first lag value of 0.729, a significant negative partial correlation at lag 2 (-0.212), and a diminishing influence at higher lags. The Q-statistics indicate significant autocorrelation with probabilities below 0.05 for the first six lags, highlighting the strong temporal dependence in the data and the need to account for these autocorrelations in modelling to ensure accurate and reliable forecasts. For separation efficiency, the correlogram of SPDMX_CM for separation efficiency shows significant ACF and PACF for 12 lags from 15 observations. The first three lags have high autocorrelation values of 0.713, 0.519, and 0.373, respectively, indicating strong correlations between data points and their preceding values. Negative correlations appear at lag 8 (-0.199) and become more pronounced at lags 9 (-0.286), 10 (-0.422), 11 (-0.382), and 12 (-0.357). The PACF shows a high first lag value of 0.713, with significant partial correlation at lag 2 (0.023) and diminishing influence at higher lags. The Q-statistics indicate significant autocorrelation with probabilities below 0.05 for the first six lags, highlighting the strong temporal dependence in the data. This necessitates accounting for these autocorrelations in modeling to ensure accurate and reliable forecasts, as indicated by the Q-Stat values such as 9.2583 (lag 1) and 53.390 (lag 12) (see, Fig. 4b).

4.3. Reliability analysis results

In this research, reliability was assessed for both fouling and separation efficiency using the Jarque-Bera test in Fig. 5. The Jarque-Bera test is utilized to statistically determine if sample data exhibit skewness and kurtosis that align with a normal distribution. It serves as a goodness-of-fit test specifically tailored for normality. The histogram represents the distribution of observations for the MX_CM membrane in a fouling study. The slight negative skewness suggests the data is slightly left-skewed, and the kurtosis indicates a platykurtic distribution. The Jarque-Bera test results, with a statistic of 1.1439 and a p-value of 0.5644, show that the data does not significantly deviate from normality, as the p-value is greater than 0.05. The PDMX_CM membranes provide skewness of −0.707231 and kurtosis of 2.434537. The negative skewness suggests the data is moderately left-skewed, and the kurtosis indicates a slightly platykurtic distribution. The Jarque-Bera test results, with a statistic of 1.4502 and a p-value of 0.4842, suggest that the data does not significantly deviate from normality, as the p-value is greater than 0.05. The histogram and distribution of SPDMX_CM membrane in fouling indicated a negative skewness, suggesting that the data is nearly symmetric, and the kurtosis indicates a distribution that is platykurtic. The Jarque-Bera test results, with a statistic of 0.5188 and a p-value of 0.7714, suggest that the data does not significantly deviate from normality, as the p-value is much greater than 0.05 (Fig. 5a).

Further analysis for separation efficiency normality indicated that the MX_CM membrane possessed high positive skewness, suggesting the data is heavily right-skewed, and the high kurtosis indicates a leptokurtic distribution, meaning the data has heavy tails. The Jarque-Bera test results, with a statistic of 15.635 and a p-value of 0.0004, indicate a significant deviation from normality, as the p-value is much less than 0.05. The PDMX_CM has high positive skewness, suggesting the data is heavily right-skewed, while the high kurtosis indicates a leptokurtic distribution, meaning the data has heavy tails. The Jarque-Bera test results, with a statistic of 37.724 and a p-value of 0.000, indicate a significant deviation from normality, as the p-value is far less than 0.05. Similarly, for the SPDMX_CM membrane in a separation efficiency, the negative skewness suggests the data is slightly left-skewed. At the same time, the kurtosis value indicates a platykurtic distribution, meaning the data has thinner tails compared to a normal distribution. The Jarque-Bera test results, with a statistic of 0.5817 and a p-value of 0.747618, indicate that the data does not significantly deviate from normality, as the p-value is much greater than 0.05 (Fig. 5b). In the context of fouling and separation efficiency studies, ensuring data normality is vital for the validity and reliability of parametric tests, such as linear regression or ANOVA. These tests assume normality, and deviations can lead to incorrect conclusions. The Jarque-Bera test combines skewness and kurtosis measures to provide a comprehensive assessment of normality, guiding researchers in choosing the correct analytical approach. In cases where data is not normally distributed, alternative methods, such as non-parametric tests or data transformations, are necessary to ensure accurate and meaningful results. This test, therefore, plays a critical role in the pre-analysis phase, ensuring robust and reliable statistical analysis in scientific research.

It is worth noting that stationarity tests are essential in data analysis to ensure that the statistical properties of the data, such as mean, variance, and autocorrelation, remain constant over time. This consistency is critical because many statistical models and forecasting techniques assume stationarity. Without it, the models may produce unreliable or biased results. By confirming stationarity, we can apply these models more effectively, leading to more accurate predictions and a better understanding of the underlying processes in the data. Table 2 presents the ADF and PP test results for three variables (MX_CM, PDMX_CM, and SPDMX_CM) to assess their stationarity in the fouling study. For MX_CM,

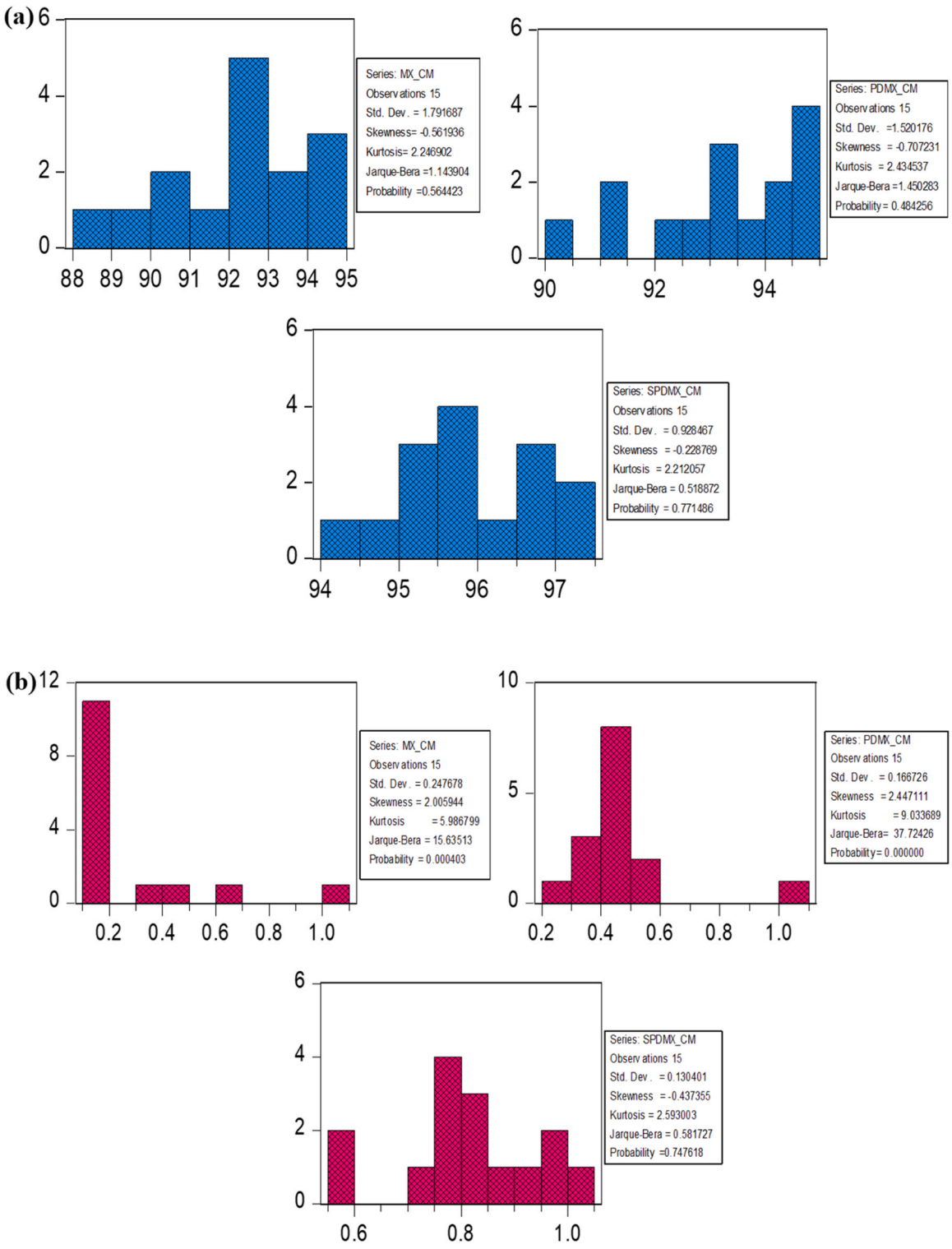


Fig. 5. Parametric analysis based on reliability and stationarity approach (a) fouling (b) separation efficiency.

ADF test values of −7.0055 and −6.1883 (both lower than the 1 % critical value of −4.122) and a PP test value of −45.884 indicate strong stationarity at level (I(0)), with significant probabilities of 0.0001 and 0.0004. PDMX_CM shows stationarity at level (I(0)) with ADF test values of −7.8419 and −23.628, but mixed results with the PP test suggesting I (1), with significant probabilities of 0.0000 and 0.0001. For SPDMX_CM, the ADF test shows non-stationarity at level with a test value of −0.3864, but stationarity after first and second differencing (I(1) and I(2)) with test values of −3.7502 and −4.225, and mixed PP test results indicating both non-stationarity and stationarity after differencing. Probabilities range from non-significant 0.8826 to significant 0.0096 (Fig. 6a). The ADF and PP tests are crucial for determining stationarity, essential for the accuracy and reliability of time series models. Stationarity ensures constant statistical properties over time, making the series stable and predictable. Ensuring data stationarity in the fouling study allows for accurate modelling, leading to more reliable predictions and analyses, and these tests help identify necessary transformations (e.g., differencing) to achieve stationarity, enhancing the robustness and validity of

Table 2
ADF and PP analysis for the fouling study.

Variables	TV	1 % CV	5 % CV	10 % CV	Prob	Decision	SC	TT
MX_CM	−7.0055	−4.122	−3.1449	−2.7138	0.0001	I (0)	AIC	ADF
	−6.1883	−4.122	−3.1449	−2.7138	0.0004	I (0)	SC	ADF
	−45.884	−4.0044	−3.0989	−2.6904	0.0001	I (0)	BK	PP
PDMX_CM	−7.8419	−4.0044	−3.0989	−2.6904	0	I (0)	AIC	ADF
	−23.628	−4.0579	−3.1199	−2.7011	0.0001	I (0)	SC	ADF
	−6.442	−4.0044	−3.0989	−2.6904	0.0002	I (1)	BK	PP
SPDMX_CM	−0.3864	−4.122	−3.1449	−2.7138	0.8826	I (0)	AIC	ADF
	−3.7502	−4.122	−3.1449	−2.7138	0.0185	I (1)	AIC	ADF
	−4.225	−4.2001	−3.1754	−2.729	0.0096	I (2)	AIC	ADF
	−0.3864	−4.122	−3.1449	−0.5995	0.8826	I (0)	SC	ADF
	−3.7502	−4.122	−3.1449	−2.7138	0.0185	I (1)	SC	ADF
	−4.225	−4.2001	−3.1754	−2.729	0.0096	I (2)	SC	ADF
	0.6442	−4.0044	−3.0989	−2.6904	0.9853	I (0)	BK	PP
	−2.5935	−4.0579	−3.1199	−2.7011	0.1188	I (1)	BK	PP
	−3.4915	−4.122	−3.1449	−2.7138	0.0283	I (2)	BK	PP

TV= test value, CV= critical value, Prob=Probability, SC= selection criteria, TT= test type, ADF=Augmented Dickey-Fuller test, PP=Phillips-Perron test statistic, Akaike information criterion (AIC), Bartlett kernel (BK), Schwarz Criteria (SC).

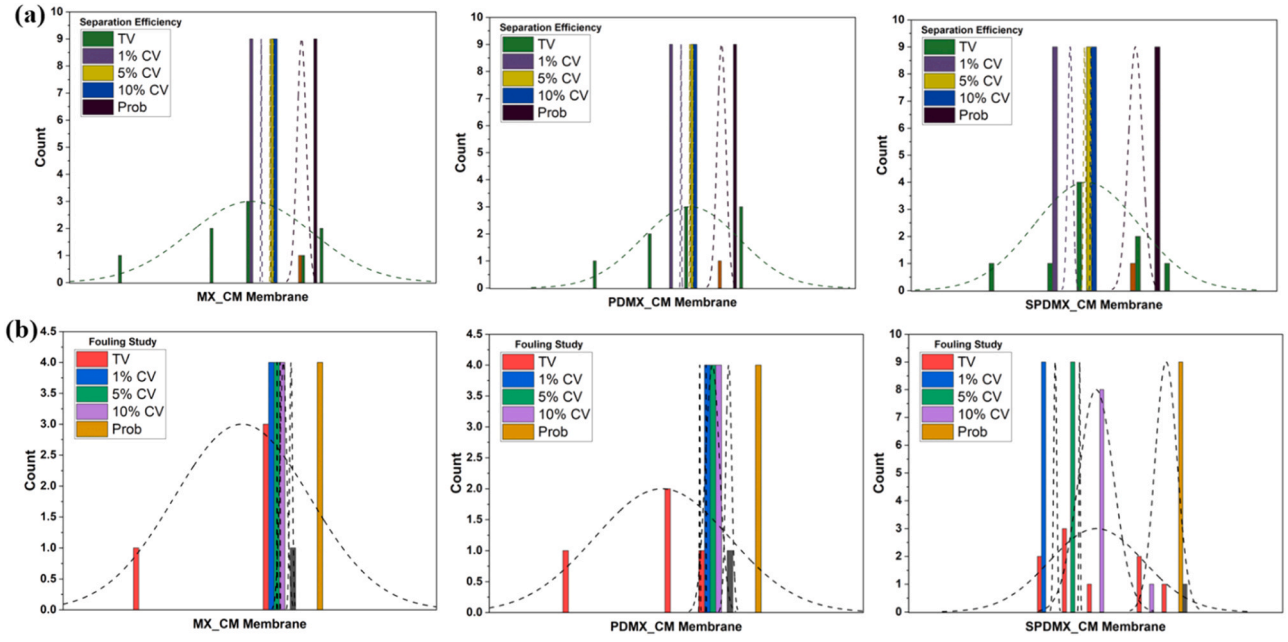


Fig. 6. Cumulative distribution plot based on probability, TV and CV for (a) fouling study (b) separation efficiency.

the study's conclusions.

Similarly, ADF and PP test results for three variables (MX_CM, PDMX_CM, and SPDMX_CM) to assess their stationarity in the separation efficiency study in Table 3. For MX_CM, ADF test values of 2.1142, −4.8288, and −9.9173 with probabilities of 0.9995, 0.0027, and 0 indicate non-stationarity at level (I(0)) but stationarity after differencing (I(1) and I(2)), supported by PP test values of 1.5162, −4.6935, and −18.268 with similar probabilities. For PDMX_CM, ADF test values of 2.5236, −2.8618, and −7.077 with probabilities of 0.9998, 0.0769, and 0.0001, along with PP test values of 3.3849, −2.9392, and −12.7782, indicate non-stationarity at level but stationarity after differencing. Also, SPDMX_CM shows ADF test values of 0.3768, −3.9426, and −3.9188 with probabilities of 0.9735, 0.0122, and 0.0198, and PP test values of 2.4545, −4.9754, and −9.5892, indicating non-stationarity at level but stationarity after differencing. These tests help identify necessary transformations, such as differencing, to achieve stationarity, enhancing the robustness and validity of the study's conclusions. The table indicates that while many series are non-stationary at their levels, they often achieve stationarity after the first or second differencing, guiding researchers inappropriately in transforming the data for further analysis.

The ADF and PP test results for the fouling and separation efficiency studies indicate varying degrees of stationarity across the analyzed variables (MX_CM, PDMX_CM, and SPDMX_CM). In the fouling study, MX_CM consistently shows stationarity at level (I(0)) with significant test values, while PDMX_CM is mostly stationary at level with occasional need for first differencing (I(1)), and SPDMX_CM requires differencing to achieve stationarity, indicated by mixed test results. In the separation efficiency study, MX_CM and PDMX_CM are non-stationary at level but become stationary after first (I (1)) or second (I(2)) differencing, as shown by both ADF and PP test values. Similarly, SPDMX_CM is non-stationary at level but achieves stationarity after differencing. Ensuring data stationarity is crucial for accurate and reliable time series modeling, particularly in ML applications. Stationary data ensures that statistical properties such as mean, variance, and autocorrelation remain constant over time, which is essential for creating stable and predictable models. In ML modeling, non-stationary data can lead to misleading results and poor model performance. Therefore, identifying and transforming non-stationary data to achieve stationarity through tests like ADF and PP is a critical preprocessing step. This process enhances the

Table 3
ADF and PP analysis for the separation efficiency study.

Variables	TV	1 % CV	5 % CV	10 % CV	Prob	Decision	SC	TT
MX_CM	2.1142	−4.0579	−3.1199	−2.7011	0.9995	I (0)	AIC	ADF
	−4.8288	−4.0579	−3.1199	−2.7011	0.0027	I (1)	AIC	ADF
	−9.9173	−4.122	−3.1449	−2.7138	0	I (2)	AIC	ADF
	2.1142	−4.0579	−3.1199	−2.7011	0.9995	I (0)	SC	ADF
	−4.8288	−4.0579	−3.1199	−2.7011	0.0027	I (1)	SC	ADF
	−9.9173	−4.122	−3.1449	−2.7138	0	I (2)	SC	ADF
	1.5162	−4.0044	−3.0989	−2.6904	0.9982	I (0)	BK	PP
	−4.6935	−4.0579	−3.1199	−2.7011	0.0034	I (1)	BK	PP
	−18.268	−4.122	−3.1449	−2.7138	0.0001	I (2)	BK	PP
PDMX_CM	2.5236	−4.0579	−3.1199	−2.7011	0.9998	I (0)	AIC	ADF
	−2.8618	−4.0579	−3.1199	−2.7011	0.0769	I (1)	AIC	ADF
	−7.077	−4.122	−3.1449	−2.7138	0.0001	I (2)	AIC	ADF
	2.3287	−4.0044	−3.0989	−2.6904	0.9998	I (0)	SC	ADF
	−2.8618	−4.0579	−3.1199	−2.7011	0.0769	I (1)	SC	ADF
	−7.077	−4.122	−3.1449	−2.7138	0.0001	I (2)	SC	ADF
	3.3849	−4.0044	−3.0989	−2.6904	1	I (0)	BK	PP
	−2.9392	−4.0579	−3.1199	−2.7011	0.0677	I (1)	BK	PP
	−12.7782	−4.122	−3.1449	−2.7138	0	I (2)	BK	PP
SPDMX_CM	0.3768	−4.0044	−3.0989	−2.6904	0.9735	I (0)	AIC	ADF
	−3.9426	−4.0579	−3.1199	−2.7011	0.0122	I (1)	AIC	ADF
	−3.9188	−4.4206	−3.2598	−2.7711	0.0198	I (2)	AIC	ADF
	0.3768	−4.0044	−3.0989	−2.6904	0.9735	I (0)	SC	ADF
	−3.9426	−4.0579	−3.1199	−2.7011	0.0122	I (1)	SC	ADF
	−3.9188	−4.4206	−3.2598	−2.7711	0.0198	I (2)	SC	ADF
	2.4545	−4.0044	−3.0989	−2.6904	0.9998	I (0)	BK	PP
	−4.9754	−4.0579	−3.1199	−2.7011	0.0022	I (1)	BK	PP
	−9.5892	−4.122	−3.1449	−2.7138	0	I (2)	BK	PP

TV= test value, CV= critical value, Prob=Probability, SC= selection criteria, TT= test type, ADF=Augmented Dickey-Fuller test, PP=Phillips-Perron test statistic, Akaike information criterion (AIC), Bartlett kernel (BK), Schwarz Criteria (SC).

robustness and validity of the models, leading to more accurate predictions and reliable analyses. The results from these tests guide researchers in applying the necessary data transformations, such as differencing, ensuring that the data meets the assumptions required for effective ML modeling.

4.4. Explainability ML results

The SHAP plot for the fouling study, as indicated in Fig. 6a, shows the impact of features (Cycle, PDMX-CM, MX-CM) on the model's output. The X-axis represents SHAP values, indicating the feature's effect on the model's predictions, with positive values showing a positive influence and negative values showing a negative influence. The Y-axis lists the features, and the color gradient (blue to red) represents low to high feature values. The plot indicates that PDMX-CM and MX-CM significantly impact the fouling performance model, with higher values of these features positively influencing the model's output. Lower values of PDMX-CM and MX-CM negatively impact the model, while the number of cycles (Cycle) has a neutral impact, suggesting it does not significantly affect the predictions. This analysis highlights the importance of PDMX-CM and MX-CM in improving fouling performance. It underscores the utility of SHAP values in enhancing model transparency and reliability, leading to better decision-making and optimization in the study. The model's prediction value is at 0.525, and the expected value is at 0.558. The Cycle feature, with a value of 0.643, positively contributes to the model's output by +0.09 units. In contrast, PDMX-CM and MX-CM, with values of 0.196 and 0.025, respectively, negatively contribute to the model's output, decreasing the prediction by −0.06 units. This indicates that the Cycle feature has a positive and significant influence on the model's prediction, while both PDMX-CM and MX-CM negatively impact the prediction to an equal extent. Understanding these contributions helps interpret the model's behavior and identify which features most significantly affect the fouling performance, enabling better decision-making and optimization in the study.

Further numerical sensitivity analysis was conducted using SHAP analysis. SHAP analysis is a method used to explain the output of ML models by attributing the contribution of each feature to the model's predictions. It is particularly valuable in our study as it provides insights into how different membrane characteristics and operational parameters influence the model's predictions on fouling and separation efficiency. By using SHAP, we can interpret the complex interactions within our models, enhancing the transparency and reliability of our findings. The SHAP plot indicates that the Cycle feature has a neutral impact on the separation efficiency model, with no significant influence on the predictions. PDMX-CM and MX-CM positively impact the model's output, particularly at higher feature values. MX-CM has the most important influence, with a noticeable spread in SHAP values indicating a strong positive impact at higher values. Understanding these contributions helps interpret the model's behavior and identify which features most significantly affect separation efficiency. This enables better decision-making and optimization in the study the SHAP with the model's prediction value at 0.551 and the expected value at 0.56. The MX-CM feature, with a value of 0.571, slightly negatively contributes to the model's output by −0.01 units. In contrast, the PDMX-CM feature, with a value of 0.624, slightly positively contributes by +0.01 units. The Cycle feature, with a value of 0.643, has a neutral impact on the model's output, contributing 0.00 units. This indicates that PDMX-CM has a slight positive influence on separation efficiency predictions, MX-CM has a slight negative influence, and Cycle has no significant effect. (Fig. 7).

The findings of this study hold significant practical implications for the application of bio-inspired MXene membranes in various real-world scenarios, particularly in industries that face the challenge of treating oil-in-water emulsions and other complex wastewater streams. The enhanced separation efficiency of these membranes ensures that they can effectively and consistently separate oil from water, making them highly suitable for use in sectors such as petrochemicals, food processing, and manufacturing, where oil-laden effluents are a common byproduct. One of the key advantages of these MXene membranes is their superior anti-fouling performance. Fouling is a major issue in membrane-based water treatment processes, as it leads to decreased efficiency, increased energy consumption, and higher operational costs

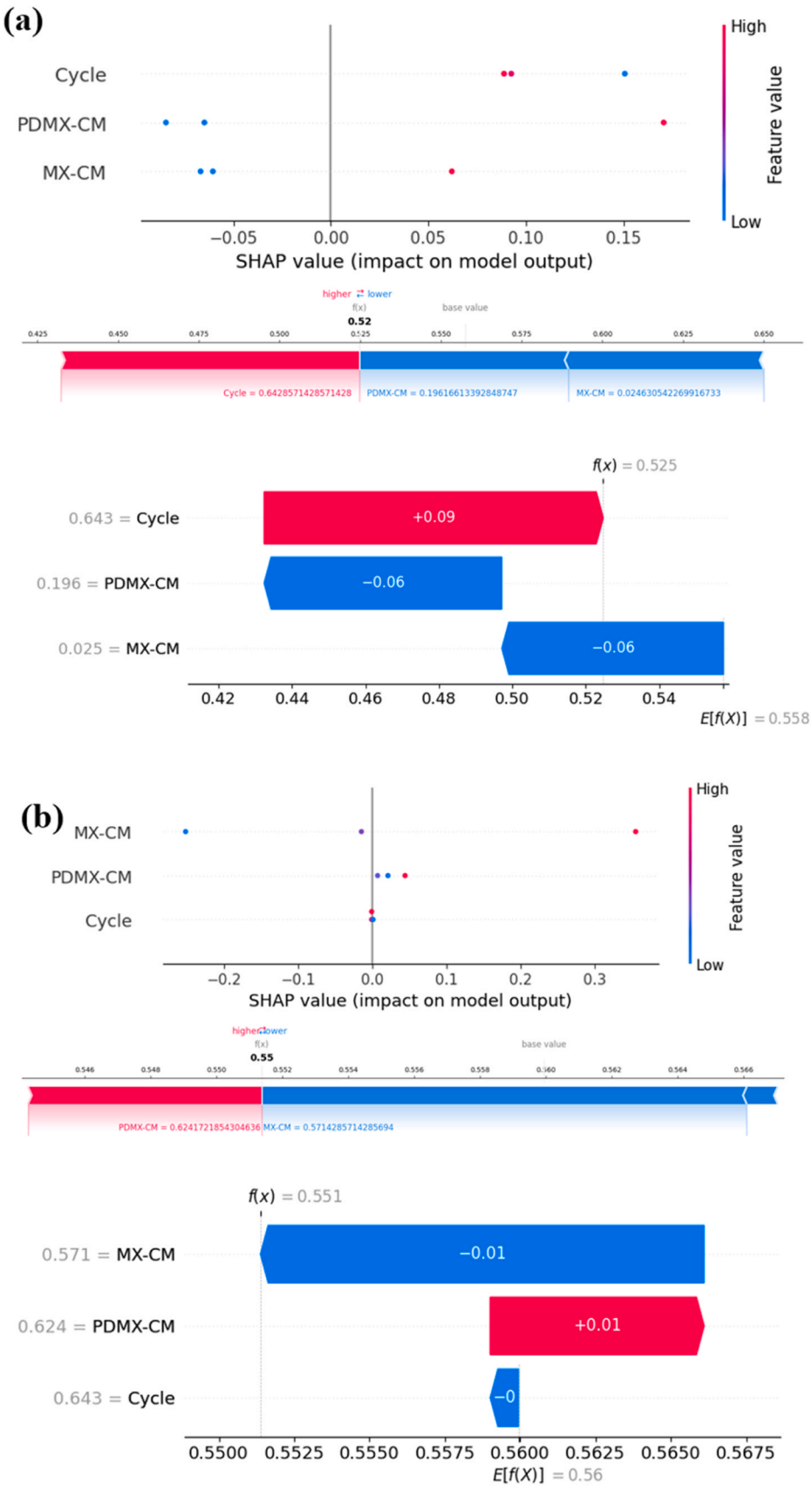


Fig. 7. SHAP explainability for (a) fouling (b) separation efficiency study.

due to the need for frequent cleaning and membrane replacement. The anti-fouling properties of these membranes allow them to maintain high performance over extended periods, significantly reducing the frequency of maintenance and replacement. This not only lowers operational costs but also minimizes downtime, which is crucial for maintaining continuous and efficient industrial operations. The state-of-the-art comparison with other literature is presented in supplementary Table S1.

Moreover, the ability of these membranes to maintain their performance in the face of fouling challenges contributes to the overall sustainability of the water treatment process. By reducing the need for harsh chemical cleaning agents and extending the lifespan of the membranes, the environmental impact of the treatment process is minimized. This aligns with broader environmental sustainability goals, particularly in industries that are increasingly focused on reducing their ecological footprint and adhering to stricter environmental regulations. Furthermore, the deployment of these membranes in industrial settings can facilitate the reuse of treated water, thereby conserving water resources and reducing the demand for fresh water in industrial processes. This is especially critical in regions facing water scarcity, where the ability to recycle and reuse water can substantially impact the environment and the economy, such as Saudi Arabia. In addition to industrial applications, these membranes could also be employed in municipal wastewater treatment plants, where they can help improve the quality of treated water and enable its reuse for non-potable applications, such as irrigation or cooling water in power plants. This can further contribute to water conservation efforts and reduce the strain on local water supplies. Generally, the application of bio-inspired MXene membranes in real-world settings offers the potential for more efficient, cost-effective, and sustainable water treatment processes. By enhancing separation efficiency, reducing fouling, and supporting water reuse, these membranes represent a valuable technology for industries seeking to improve their operational efficiency while also advancing their environmental stewardship and sustainability goals.

5. Conclusion

This study evaluated the fouling and separation efficiency of MX-CM, PDMX-CM, and SPDMX-CM membranes using a combination of statistical tests and machine learning (ML) models. The results demonstrated that ensuring data stationarity is critical for reliable ML modeling. MX-CM consistently showed stable performance data, making it a reliable candidate for further analysis, while PDMX-CM and SPDMX-CM required preprocessing steps like differencing to achieve stationarity. SHAP (Shapley Additive Explanations) analysis provided valuable insights into the influence of different features on model predictions. In the fouling study, higher values of PDMX-CM and MX-CM positively impacted predictions, with Cycle having a moderate positive effect. In contrast, in the separation efficiency study, Cycle had a neutral impact, while PDMX-CM and MX-CM showed slight differential effects. The study underscores the importance of rigorous data preprocessing and the use of explainable ML techniques to optimize membrane performance in water treatment. These findings are crucial for advancing membrane technology, supporting environmental sustainability, and aligning with global standards like the Sustainable Development Goals (SDGs) and Environmental Protection Agency (EPA) guidelines. The key findings are summarized as follow:

- Ensuring data stationarity is crucial for reliable ML modeling. MX-CM shows stable performance data and requires no difference, while PDMX-CM and SPDMX-CM need preprocessing.
- MX-CM membrane demonstrated consistent stability, making it a reliable option for further analysis, while PDMX-CM and SPDMX-CM required additional steps to achieve similar reliability.
- SHAP analysis revealed that higher values of PDMX-CM and MX-CM positively influenced fouling predictions, with Cycle having a moderate positive impact.
- In the separation efficiency study, SHAP analysis indicated that Cycle had a neutral impact, while PDMX-CM and MX-CM showed slight but differential effects on model predictions.
- Accurate data preprocessing, including differencing, is necessary to ensure the robustness of time-series models used in membrane performance studies.
- The findings support optimizing membrane performance in water treatment, contributing to environmental sustainability and aligning with global standards like the SDGs and EPA guidelines.
- Based on our findings, future research could explore the development of advanced pre-processing techniques to further enhance the stationarity and reliability of membrane performance data, particularly for PDMX-CM and SPDMX-CM membranes.
- Moreover, future studies could investigate the application of SHAP analysis in real-time monitoring and optimization of membrane processes, aiming to improve predictive accuracy and operational efficiency in industrial water treatment settings.

CRediT authorship contribution statement

Nadeem Baig: Writing – original draft, Resources, Methodology, Conceptualization. **Sani Isah Abba:** Writing – review & editing, Writing – original draft, Visualization, Validation, Conceptualization. **Jamil Usman:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Ibrahim Muhammad:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Ismail Abdulazeez:** Validation, Supervision, Software, Project administration, Conceptualization. **A.G. Usman:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Isam H. Aljundi:** Writing – review & editing, Validation, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgements

The authors would like to acknowledge all support from the Interdisciplinary Research Center for Membranes and Water Security, King Fahd University of Petroleum and Minerals.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.clwat.2024.100041.

References

Al-Hamadani, Y.A.J., et al., 2020. Applications of MXene-based membranes in water purification: a review. Chemosphere vol. 254, 126821. https://doi.org/10.1016/J.CHEMOSPHERE.2020.126821.

Anjum, M., Khan, K., Ahmad, W., Ahmad, A., Amin, M.N., Nafees, A., 2022. New shapley additive explanations (SHAP) approach to evaluate the raw materials interactions of steel-fiber-reinforced concrete. Materials vol. 15 (18). https://doi.org/10.3390/ma15186261.

A.B. Arrieta *et al.*, "Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI".2024.

Baig, N., et al., 2023c. Highly permeable sulfonated polydopamine integrated MXene membranes for efficient surfactant-stabilized oil-in-water separation. Langmuir vol. 39 (39), 13953–13967. https://doi.org/10.1021/acs.langmuir.3c01651.

Baig, N., 2023. Two-dimensional nanomaterials: a critical review of recent progress, properties, applications, and future directions. Compos Part A Appl. Sci. Manuf. vol. 165, 107362 https://doi.org/10.1016/j.compositesa.2022.107362.

Baig, N., Abdulazeez, I., Aljundi, I.H., 2023a. Low-pressure-driven special wettable graphene oxide-based membrane for efficient separation of water-in-oil emulsions. *2023 6:1* NPJ Clean. Water vol. 6 (1), 1–13. https://doi.org/10.1038/s41545-023-00252-y.

Baig, N., Abdulazeez, I., Aljundi, I.H., 2023b. Low-pressure-driven special wettable graphene oxide-based membrane for efficient separation of water-in-oil emulsions. *2023 6:1* npj Clean. Water vol. 6 (1), 1–13. https://doi.org/10.1038/s41545-023-00252-y.

Cao, Z., Barati Farimani, O., Ock, J., Barati Farimani, A., 2023. Machine learning in membrane design: from property prediction to AI-Guided Optimization. Nano Lett. vol. 20, 23. https://doi.org/10.1021/ACS.NANOLETT.3C05137/ASSET/IMAGES/LARGE/NL3C05137_0004.JPEG.

Chen, Z., Yam, V.W.W., 2020. Precise size-selective sieving of nanoparticles using a highly oriented two-dimensional supramolecular polymer. Angew. Chem. Int. Ed. vol. 59 (12), 4840–4845. https://doi.org/10.1002/ANIE.201913621.

Cheng, L., Liu, G., Zhao, J., Jin, W., 2021. Two-dimensional-material membranes: manipulating the transport pathway for molecular separation. Acc. Mater. Res vol. 2 (2), 114–128. https://doi.org/10.1021/ACCOUNTSMR.0C00092/SUPPL_FILE/MR0C00092_SI_001.PDF.

Chertopalov, S., Mochalin, V.N., 2018b. Environment-sensitive photoresponse of spontaneously partially oxidized Ti3C2 MXene thin films. ACS Nano vol. 12 (6), 6109–6116. https://doi.org/10.1021/ACSNANO.8B02379/SUPPL_FILE/NN8B02379_SI_001.PDF.

Chertopalov, S., Mochalin, V.N., 2018a. Environment-sensitive photoresponse of spontaneously partially oxidized Ti3C2 MXene thin films. ACS Nano vol. 12 (6), 6109–6116. https://doi.org/10.1021/ACSNANO.8B02379/ASSET/IMAGES/LARGE/NN-2018-02379Y_0007.JPEG.

T. Fawcett, "Data Science for Business," no. August 2013, 2019.

Frey, N.C., Wang, J., Vega Bellido, G.I., Anasori, B., Gogotsi, Y., Shenoy, V.B., 2019b. Prediction of synthesis of 2D Metal Carbides and Nitrides (MXenes) and their precursors with positive and unlabeled machine learning. ACS Nano vol. 13 (3), 3031–3041. https://doi.org/10.1021/acsnano.8b08014.

Frey, N.C., Wang, J., Vega Bellido, G.I., Anasori, B., Gogotsi, Y., Shenoy, V.B., 2019a. Prediction of synthesis of 2D Metal Carbides and Nitrides (MXenes) and their precursors with positive and unlabeled machine learning. ACS Nano vol. 13 (3), 3031–3041. https://doi.org/10.1021/ACSNANO.8B08014/ASSET/IMAGES/MEDIUM/NN-2018-08014V_M023.GIF.

Hamisu, M.A., Salleh, K.M., Sern, L.C., Adamu, B.Y., Gambo, K., 2017. Proposed competency model for technical and vocational education and training (TVET) lecturers teaching in technical colleges, bauchi state in perspective. Path Sci. vol. 3 (9), 5001–5009. https://doi.org/10.22178/pos.26-7.

Iban, M.C., 2022. An explainable model for the mass appraisal of residences: the application of tree-based Machine Learning algorithms and interpretation of value determinants. Habitat Int. vol. 128 (August), 102660 https://doi.org/10.1016/j.habitatint.2022.102660.

Jahani, H., Jain, R., Ivanov, D., 2023. Data science and big data analytics: a systematic review of methodologies used in the supply chain and logistics research. Ann. Oper. Res. https://doi.org/10.1007/s10479-023-05390-7.

Kim, J., et al., 2021. Chemically stabilized and functionalized 2d-mxene with deep eutectic solvents as versatile dispersion medium. Adv. Funct. Mater. vol. 31 (13), 2008722. https://doi.org/10.1002/ADFM.202008722.

Kwon, O., et al., Jan. 2022. A comprehensive review of MXene-based water-treatment membranes and technologies: recent progress and perspectives. Desalination vol. 522, 115448. https://doi.org/10.1016/J.DESAL.2021.115448.

A. Lehnert, F. Gawantka, J. During, and F. Just, "XplAInable: Explainable AI Smoke Detection at the Edge," pp. 1–22, 2024.

Liang, J., Zhu, X., 2019. Phillips-inspired machine learning for band gap and exciton binding energy prediction. J. Phys. Chem. Lett. vol. 10 (18), 5640–5646. https://doi.org/10.1021/acs.jpclett.9b02232.

Liu, P., Hou, J., Zhang, Y., Li, L., Lu, X., Tang, Z., 2020. Two-dimensional material membranes for critical separations (Jun.). Inorg. Chem. Front vol. 7 (13), 2560–2581. https://doi.org/10.1039/D0QI00307G.

Lu, Y., Fan, X., Zhang, Y., Wang, Y., Jiang, X., 2023. Machine Learning Models Using SHapley Additive exPlanation for Fire Risk Assessment Mode and Effects Analysis of Stadiums. Sensors vol. 23 (4). https://doi.org/10.3390/s23042151.

S.M. Lundberg and S. Lee, "A Unified Approach to Interpreting Model Predictions," no. November, 2017.

Mati, S., 2021. Do as your neighbours do? Assessing the impact of lockdown and reopening on the active COVID-19 cases in Nigeria. Soc. Sci. Med. vol. 270 (2020), 113645 https://doi.org/10.1016/j.socscimed.2020.113645.

Paechotrattanakul, P., Jitapunkul, K., Iamprasertkun, P., Srinoi, P., Sirisaksoontorn, W., Hirunpinyopas, W., 2023. Ultrahigh stable laminar graphene membranes for effective ionic and molecular nanofiltration with a machine learning-assisted study. Nanoscale vol. 15 (19), 8716–8729. https://doi.org/10.1039/D2NR06969E.

Priya, P., Nguyen, T.C., Saxena, A., Aluru, N.R., 2022. Machine learning assisted screening of two-dimensional materials for water desalination. ACS Nano vol. 16 (2), 1929–1939. https://doi.org/10.1021/ACSNANO.1C05345/ASSET/IMAGES/MEDIUM/NN1C05345_M007.GIF.

Qian, C., Sun, K., Bao, W., 2022. Recent advance on machine learning of MXenes for energy storage and conversion. Int J. Energy Res vol. 46 (15), 21511–21522. https://doi.org/10.1002/ER.7833.

Rodríguez-Pérez, R., Bajorath, J., 2020. Interpretation of compound activity predictions from complex machine learning models using local approximations and shapley values. J. Med. Chem. vol. 63 (16), 8761–8777. https://doi.org/10.1021/acs.jmedchem.9b01101.

Shuabiu, U.A., Usman, M.A.M., Çavuşoğlu, B., 2021. The nexus among competitively valued exchange rates, price level, and growth performance in the Turkish Economy; new insight from the global value chains. J. Risk Financ. Manag. vol. 14 (11), 528. https://doi.org/10.3390/jrfm14110528.

Sutrisna, P.D., Kurnia, K.A., Siagian, U.W.R., Ismadji, S., Wenten, I.G., 2022. Membrane fouling and fouling mitigation in oil–water separation: a review. J. Environ. Chem. Eng. vol. 10 (3), 107532 https://doi.org/10.1016/J.JECE.2022.107532.

Tummons, E., Han, Q., Tanudjaja, H.J., Hejase, C.A., Chew, J.W., Tarabara, V.V., 2020. Membrane fouling by emulsified oil: A review (Oct.). Sep Purif. Technol. vol. 248, 116919. https://doi.org/10.1016/J.SEPPUR.2020.116919.

Usman, J., Abba, S.I., Baig, N., Abu-Zahra, N., Hasan, S.W., Aljundi, I.H., 2024. Design and machine learning prediction of in situ grown PDA-Stabilized MOF (UiO-66-NH2) Membrane for Low-Pressure Separation of Emulsified Oily Wastewater. ACS Appl. Mater. Interfaces. https://doi.org/10.1021/ACSAMI.4C00752.

Wei, W., Liu, J., Jiang, J., 2019. "Computational Design of 2D covalent-organic framework membranes for organic solvent nanofiltration. ACS Sustain Chem. Eng. vol. 7 (1), 1734–1744. https://doi.org/10.1021/ACSSUSCHEMENG.8B05599/SUPPL_FILE/SC8B05599_SI_003.AVI.

G.T. Wilson, "Time Series Analysis: Forecasting and Control, 5th Edition, by George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel and Greta M. Ljung, 2015. Published by John Wiley and Sons Inc., Hoboken, New Jersey, pp. 712. ISBN: 978-1-118-67502-1," Journal of Time Series Analysis, vol. 37, no. 5, pp. 709–711, 2016, doi: 10.1111/jtsa.12194.

Xu, P., Ji, X., Li, M., Lu, W., 2023. Small data machine learning in materials science. NPJ Comput. Mater. vol. 9 (1), 1–15. https://doi.org/10.1038/s41524-023-01000-z.

Yan, F., Song, K., Liu, Y., Chen, S., Chen, J., 2020. Predictions and mechanism analyses of the fatigue strength of steel based on machine learning. J. Mater. Sci. vol. 55 (31), 15334–15349. https://doi.org/10.1007/s10853-020-05091-7.

Ying, Y., Yang, Y., Ying, W., Peng, X., 2016. Two-dimensional materials for novel liquid separation membranes. Nanotechnology vol. 27 (33), 332001. https://doi.org/10.1088/0957-4484/27/33/332001.

Yu, H., Wen, X., Wu, M., Sheng, D., Wu, J., Zhao, Y., 2022. Data-based groundwater quality estimation and uncertainty analysis for irrigation agriculture. Agric. Water Manag. vol. 262 (320), 107423 https://doi.org/10.1016/j.agwat.2021.107423.

Zhang, W., Zhang, L., Zhao, H., Li, B., Ma, H., 2018. A two-dimensional cationic covalent organic framework membrane for selective molecular sieving. J. Mater. Chem. A Mater. vol. 6 (27), 13331–13339. https://doi.org/10.1039/C8TA04178D.